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Transaction card assembly

Abstract

A modular transaction card assembly includes a card frame having the traditional dimensions of a credit card, and a transaction card that is smaller than a traditional card and that fits into a receptacle of the card frame. Each of the card frame and the transaction card may be capable of performing contactless data transactions individually. In some instances, the combined assembly of the card frame with the transaction card may detect a proximity of a mobile device to the assembly and generate an authentication credential that is unique to the combination of the card frame, the transaction card, and the mobile device. The authentication credential may be used to authentic the transaction card when conducting a data transaction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/829,761, filed Jun. 1, 2022, which is a continuation-in-part of U.S. application Ser. No. 17,539,633, filed on Dec. 1, 2021, and a continuation-in-part of U.S. application Ser. No. 17/539,636, filed on Dec. 1, 2021, which are hereby incorporated by reference in its entirety.

FIELD OF USE
(1) Aspects of the disclosure relate generally to transaction cards and more specifically to a modular transaction card and authentication of the modular transaction card when used in data transactions.
BACKGROUND
(2) Transaction cards have different form factors with different capabilities. Traditional credit cards, for example, may perform transactions contactlessly, using a magnetic strip, or via a smart chip. Transaction cards in smaller form factors, such as ones with a hole punch that can be attached to a keychain, are often more convenient to carry and may have the same transaction features, but not all card readers are able to work with the smaller dimensions.
(3) Moreover, because such transaction cards are highly portable and can operate wirelessly, they are susceptible to being “skimmed,” where a person with a hand held scanner surreptitiously copies or reads information from the transaction card by placing a hand held contactless card reader near the transaction card.
SUMMARY
(4) The following presents a simplified summary of various aspects described herein. This summary is not an extensive overview, and is not intended to identify key or critical elements or to delineate the scope of the claims. The following summary merely presents some concepts in a simplified form as an introductory prelude to the more detailed description provided below.
(5) The methods, devices, systems, and/or computer-readable media disclosed herein relate to a transaction card assembly that includes a card frame having the traditional dimensions of a credit card, and a transaction card that is smaller than a traditional credit card (e.g., a “mini” transaction card). The transaction card may be capable of performing contactless data transactions (for example, credit card purchases) on its own via wireless communications, but may not be compatible with certain card readers, such as a chip reader, which accepts only a traditional credit card format. The card frame includes a receptacle that accepts and secures the transaction card and couples it to an antenna in the card frame, permitting the combined card frame and transaction card assembly to perform data transactions as a traditional form factor transaction card.
(6) The receptacle in the card frame may have electrical contacts that mate with electrical contacts on the transaction card to couple it to the card frame antenna. In some variations, the card frame may have a second antenna that wirelessly communicates with an antenna in the transaction card when it is secured in the receptacle.
(7) The card frame may include other features, such as electronics to provide power to the transaction card in the receptacle and may include shielding to prevent the antenna in the transaction card and the antenna in the card frame from both communicating with a card reader simultaneously.
(8) In some variations, the card frame comprises a processing circuit for implementing a smart card frame. The processing circuit may include a computing device and memory storing computer instructions for enabling the card frame to work on its own as a transaction card without the smaller transaction card in the receptacle. The computing device may further implement authentication and cryptographic functions. For example, the card frame may authenticate the transaction card in the receptacle as a condition for completing a data transaction. The card frame may be used with multiple different transaction cards, with each combination of the card frame and different transaction cards having a unique identifier that is distinct from the identifier of each transaction card alone and different from the identifier of the card frame alone.
(9) Methods, devices, and systems disclosed herein also provide further security measures through the use of a mobile device in conjunction with the smart card frame and transaction card to authenticate the transaction card. Proximity of the mobile device to the transaction card or card frame may be taken into account when generating a unique identifier or credential for authenticating the transaction card or authorizing a transaction

using the transaction card.
(10) These features, along with many others, are discussed in greater detail below.

Description

DESCRIPTION OF THE DRAWINGS
(1) The present disclosure is described by way of example and not limited in the accompanying figures in which like reference numerals indicate similar elements and in which:
(2) FIG. **1** illustrates a system in which a transaction card assembly may be used in accordance with one or more aspects of the disclosure:
(3) FIGS. **2**A-**2**H illustrate multiple views of a transaction card assembly in accordance with one or more aspects of the disclosure:
(4) FIGS. **3**A-**3**D illustrate multiple views of a transaction card assembly device in accordance with one or more aspects of the disclosure:
(5) FIG. **4** illustrates a block diagram of an electrical circuit according to one or more aspects of the disclosure:
(6) FIG. **5** illustrates a first example method for using a transaction card assembly to perform a data transaction according to one or more aspects of the disclosure.
(7) FIG. **6** illustrates a second example method for using a transaction card assembly to perform a data transaction according to one or more aspects of the disclosure.
(8) FIG. **7** illustrates a block diagram of a processing circuit according to one or more aspects of the disclosure:
(9) FIG. **8** illustrates a flow chart of a process for using a transaction card assembly to perform a data transaction according to one or more aspects of the disclosure:
(10) FIG. **9** illustrates a system in which a transaction card assembly and a mobile device may be used in accordance with one or more aspects of the disclosure:
(11) FIG. **10** illustrates a flow chart of a process for using a transaction card assembly and a mobile device to perform a data transaction according to one or more aspects of the disclosure.
DETAILED DESCRIPTION
(12) In the following description of the various embodiments, reference is made to the accompanying drawings, which form a part hereof, and in which is shown by way of illustration various embodiments in which aspects of the disclosure may be practiced. It is to be understood that other embodiments may be utilized and structural and functional modifications may be made without departing from the scope of the present disclosure. Aspects of the disclosure are capable of other embodiments and of being practiced or being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. Rather, the phrases and terms used herein are to be given their broadest interpretation and meaning. The use of “including” and “comprising” and variations thereof is meant to encompass the items listed thereafter and equivalents thereof as well as additional items and equivalents thereof.
(13) By way of introduction, aspects discussed herein may relate to components, methods and techniques for a smart card system for performing contactless data transactions, the smart card system. The smart card system includes a smart card frame and one or more transaction cards. Each transaction card includes a first antenna and a first processor circuit that may perform, via the first antenna, a first contactless data transaction with a terminal. The first contactless data transaction may be based on a first secure credential stored in and unique to that transaction card. The smart card frame includes a flat sheet and a receptacle integrated in the flat sheet. The receptacle may be configured to secure, enable removal of, and resecure each of the one or more transaction cards in the smart card frame. The smart card frame may include a second antenna; and a second processor circuit integrated in the flat sheet. For each of the one or more transaction cards, the second processor circuit may be configured to generate a third secure credential based on a second secure credential stored in the smart card frame and based on the first secure credential of the transaction card. The third secure credential may be unique to the combination of the transaction card and the smart card frame. The combined smart card frame and a transaction card may then perform a second contactless data transaction with the terminal based on the third secure credential.
(14) Aspects discussed herein may also relate to systems, methods, and techniques for a smart card frame and transaction card to generate a secure credential based on whether a paired mobile device is in the vicinity of

the transaction card when a data transaction is being performed. The secure credential may be unique the combination of the transaction card and the mobile device, or unique to the combination of the card frame, transaction card, and the mobile device. The transaction card, or the combined smart card frame/transaction card may then perform a data transaction with a terminal based on the secure credential.
(15) Further aspects relate to different combinations of a plurality of transaction cards, a plurality of card frames, and a plurality of mobile devices. Each of the plurality of card frames, each of the plurality of transaction cards, and each of the plurality of mobile devices respectively may have one of a plurality of unique credentials stored therein. Each combination of a first credential (from one of the plurality of transaction cards), a second credential (from one of the plurality of card frames), and a third credential (from one of the plurality of mobile devices) generates a unique authentication credential for performing respective data transactions between the transaction cards and a terminal.
(16) FIG. **1** illustrates a system **10** that illustrates several components that may be found when conducting a transaction with different types of transaction cards. For example, system **10** shows a card reader **120** (e.g., a point-of-sale terminal), that may exchange data with transaction cards (e.g., **102**) through a plurality of communication techniques. The card reader **120** may be communicatively coupled to a server **140** via network **130**.
(17) Card reader **120** may be any suitable card reader capable of exchanging data and/or information with transaction cards **102**. In this regard, card reader **120** may be a chip-based reader, a magnetic-based reader, an EMV reader, a wireless based reader, or any combination thereof. Accordingly, card reader **120** may include a display, a keypad, a network interface and a card interface. The display may present information to the cardholder, such as the amount owed, the status of the transaction, and whether the transaction has been approved or denied. A keypad or touch screen may allow a cardholder to input a personal identification number (PIN) code, password, an amount for withdrawal, and the like. A network interface may be a wired connection, wireless connection, a short-range wireless connection, a near field communication (NFC) connection, or any combination thereof. The network interface may permit card reader **120** to communicate with server **140**, via network **130**, for example, to authorize a transaction. The card interface may permit card reader **120** to communicate with transaction cards **102**. In these instances, card reader **120** may convey information related to the cardholder's account to transaction cards **102**. Card reader **120** may be limited in the ways it can communicate with different types of transaction cards. For example, card reader **120** may have a transaction card chip reader that only works with the dimensions of a standard size credit card, but not with the dimensions of transaction card **102**, which in some instances, may have smaller or non-standard dimensions (e.g., a mini-card connectable to a key ring).
(18) Various aspects described herein, which address this compatibility issue, are directed to a card assembly **100** comprising a card frame **101** that secures a smaller transaction card **102** in a receptacle **103**. In some embodiments, the card assembly **100** performs data transactions with card readers not compatible with transaction card **102**, and optionally, with a unique identifier that is distinct from an identifier of transaction card **102** when the transaction card is not secured in the card frame **101**.
(19) Server **140** may be a stand-alone server, a corporate server, or a server located in a server farm or cloud-computer environment. According to some examples, server **140** may be a virtual server hosted on hardware capable of supporting a plurality of virtual servers. Server **140** may be configured to execute server-based software configured to provide cardholders with access to account information and perform routing banking functions. In some embodiments, the server-based software corresponds to client-based software executing on card reader **120**.
(20) Network **130** may be any type of communications and/or computer network. The network **130** may include any type of communication mediums and/or may be based on any type of communication standards or protocols. In this regard, network **130** may include the Internet, a local area network (LAN), a wide area network (WAN), a wireless telecommunications network, and/or any other communication network or combination thereof.
(21) Devices and systems **120**, **130**, and **140** in FIG. **1** may be implemented, in whole or in part, using one or more computing systems, for example, as described below with respect to FIG. **7**.
(22) Transaction card **102** and card frame **101**, individually or combined as card assembly **100**, may be configured to permit a cardholder to access one or more types of accounts. In this regard, transaction card **102**, card frame **101**, and/or card assembly **100** may behave as a credit card, a charge card, a debit card, a prepaid card, a smartcard, a payment card or an EMV card. In some embodiments, transaction card **102**, card frame **101**, and/or card assembly **100** may be an identification card, a club membership card, a rail pass card, or a

building access card. As will be discussed in greater detail with respect to FIGS. **2** and **3**, transaction card **102**, card frame **101**, and/or card assembly **100** may be chip-enabled and/or may include a magnetic strip. In further embodiments, transaction card **102**, card frame **101**, and/or card assembly **100** may include NFC capabilities, short-range wireless communication capabilities (e.g., Bluetooth®), wireless communication capabilities (e.g., Wi-Fi), or any combination thereof. The NFC capabilities, short-range wireless communication capabilities, and wireless communication capabilities may be referred to collectively as communication capabilities. These communication capabilities may permit transaction card **102**, card frame **101**, and/or card assembly **100** to communicate with card reader **120**.

(23) Turning to FIG. **2**A-**2**H, various views of card assembly **100** are illustrated. As illustrated in FIGS. **2**A and **2**B illustrating front and back views, card frame **101** of card assembly **100** may be of a standard size and made of a suitable substrate, such as plastic, metal, etc. For example, card frame **101** may be formed as a flat sheet having a rounded rectangle perimeter. In some variations, the flat sheet is 3.361 to 3.382 inches wide, 2.119 to 2.133 inches high, and 0.027 to 0.033 inches thick.

(24) Card frame **101** may include a magnetic strip **204** for storing data (e.g., credit card information) that may be read and written to by card reader **120**, and may include an antenna **201** capable of wireless communications (e.g., NFC, Bluetooth, Wi-Fi) with another device, such as card reader **120** in FIG. **1**.

(25) Receptacle **103** in card frame **101** may be configured to secure, permit removal of, and resecure transaction card **102**. Card frame **101** may further include an electric circuit **205** for interfacing transaction card **102** to card frame antenna **201**, when transaction card **102** is secured in the receptacle **103**. Details of electric circuit **205** are further described below with respect to FIGS. **4** and **7**.

(26) Transaction card **102** may include a computer chip **203** and its own antenna **202** capable of wireless communications (e.g., NFC, Bluetooth, Wi-Fi) with another device, such as carder reader **120** in FIG. **1**. As illustrated in the figures, antenna **202** and computer chip **203** may be embedded within transaction card **102**, and may be located anywhere in the perimeter of the transaction card **102** as illustrated in FIGS. **2**A and **2**B, and at any depth or on either surface of the transaction card **102**.

(27) FIG. **2**C illustrates a cross-sectional view A-A of FIG. **2**B. As illustrated in FIG. **2**C, the magnetic strip **204** may be embedded in, or on the surface of one side of, card frame **101**, enabling it to be read by a magnetic strip card reader. Antenna **201** and electric circuit **205** may also be embedded in, or on the surface of one side of card frame **101**. As illustrated in FIGS. **2**A-**2**C, antenna **201** may be integrated along the perimeter of card frame **101**. In some variations, card frame **101** comprises a flat metal sheet and the antenna **201** is insulated from the metal sheet. In further variations, the antenna **201** is exposed along the edge of the card frame or is not completely surrounded by the metal sheet, so that the metal sheet does not interfere with reception and transmission of radio frequency communications by the antenna **201**.

(28) The locations of magnetic strip **204**, antenna **201**, and electric circuit **205** are not limited to those locations illustrated in the figures and may be located anywhere in the perimeter of the card frame **101** illustrated in FIGS. **2**A and **2**B, and at any depth or on either surface of the card frame **101** illustrated in FIG. **2**C.

(29) As illustrated in FIGS. **2**A-**2**C, in some variations transaction card **102**, when secured in receptacle **103**, may be substantially within the outer dimensions of card frame **101**, such that card frame **101** and transaction card **102** together form card assembly **100** as a uniform piece that appears and functions as a traditional transaction card (e.g., credit card).

(30) Computer chip **203** in transaction card **102** may be a smart chip or integrated circuit. In this regard, chip **203** may include a microprocessor and memory, such as read only memory (ROM) and/or random access memory (RAM). Additionally, chip **203** may include one or more contact pads (illustrated in FIG. **2**A) to receive electric power to operate the transaction card **102** and exchange signals with a terminal, such as card reader **120**. In some instances, the chip **203** may be configured to execute one or more applications. The applications may allow chip **203** to process payments. In other examples, the applications may allow the chip **203** to perform cryptographic processing, authentication, define risk management parameters (e.g., when the transaction may be conducted offline), digitally sign payment data, and/or verify the cardholder. When secured in the receptacle **103** of the card frame **101**, the contact pads of chip **203** may be positioned to appear as contact pads for the card frame **101**. In some variations, transaction card **102** may be configured to perform a first transaction (e.g., a data transaction via chip **203** and/or antenna **202** authenticated with a first set of credentials) when the transaction card **102** is not inserted into card frame **101**, and perform a second transaction (e.g., a data transaction via chip **203** and/or antenna **201** authenticated with a second set of credentials) when the transaction card **102** is inserted into card frame **101**. In some variations, chip **203** may

be implemented with computing device **700** as illustrated in FIG. **7**, which is further described below.
(31) FIGS. **2**D-**2**F illustrate various examples of view B of FIG. **2**C to show in more detail the transaction card **102** inserted in receptacle **103**. As illustrated in FIGS. **2**D and **2**E, card frame **101** may be a flat sheet comprising two opposing surfaces separated by a thickness and bounded by a perimeter (illustrated in FIGS. **1**A and **1**B), wherein the receptacle **103** comprises a hole passing completely through the thickness of the flat sheet. The receptacle **103** may have a perimeter (as illustrated in FIGS. **2**A and **2**B) that matches entirely, or only at some edges of, the perimeter of the transaction card **102**. As illustrated in FIG. **2**D, the transaction card **102** and receptacle **103** may have an interference fit in which the profile of the transaction card **102** has a protrusion **208**, which fits within a groove **207** of the receptacle **103** to secure the transaction card **102** in the card frame **101**. The material of the protrusion **208** and/or the walls of the groove **207** may be flexible to allow the transaction card **102** to be secured, removed, and resecured in the receptacle **103** with the application of opposing forces perpendicular to the faces of the card frame **101** and transaction card **102** respectively, for example, to snap the transaction card **102** into the receptacle **103**. While protrusion **208** is illustrated as curved and groove **2007** is illustrated in the shape of a "v," these may be of any profile that provides an interference fit. Additionally, the profiles may be reversed so that the perimeter of the transaction card **102** has a groove, and the receptacle **103** has a protrusion.
(32) FIG. **2**E illustrates another example, in which the perimeters of the receptacle **103** and transaction card **102** have mating rims **210** and **209** respectively with mirrored profiles. In some variations, the transaction card **102** may be inserted from only one side of the card frame **101**. The transaction card **102** may be secured in the receptacle **103** by friction between the perimeters of the transaction card **102** and receptacle **103**. In other variations, the transaction card **102** may be magnetically coupled to the receptacle **103** and/or card frame **101** to secure, enable removal of, and resecure the transaction card **102** in the card frame **101**. Each of these configurations can be used together, for example by some edges of the transaction card **102** being secured with a groove/protrusion, and some with mirrored rims that are coupled magnetically. Transaction card **102** may be configured to detect whether it is inserted in the card frame **101**, and based on this detection, perform different operations (e.g., perform different types of data transactions or take on different identities).
(33) While FIGS. **2**D and **2**E illustrate the receptacle **103** as a hole passing completely through the flat sheet of the card frame, in other variations, the receptacle **103** may be a recess in one of the two opposing surfaces with the other surface being completely or partially closed.
(34) FIG. **2**F illustrates another variation of receptacle **103** comprising a slot **211** having an open end, along the perimeter between the two opposing surfaces of the card frame **101**, through which the receptacle **103** is configured to secure, enable the removal of, and resecure the transaction card **102**.
(35) In some variations, the card frame **101** and/or receptacle **103** may provide an electromagnetic shield preventing the antenna **202** in the transaction card **102** from receiving or transmitting radio frequency signals while the transaction card **102** is secured in the receptacle **103**. For example, in the receptacle **103** in FIG. **2**F, one or both surfaces of the card frame **101** may be coated or made of a conductive material (e.g., aluminum, stainless steel, titanium), which covers the transaction card **102** partially or completely. In this way, the card frame **101** may disable the transaction card antenna **202** while the transaction card **102** is secured in the receptacle **103**, thus preventing both antennas **201** and **202** from relaying data transactions simultaneously. Alternatively or additionally, transaction card **102** may disable its antenna **202** based on detecting that the transaction card **102** is secured in the receptacle **103**.
(36) As previously discussed, when transaction card **102** is secured in receptacle **103**, it may be interfaced to antenna **201** in the card frame **101** via electric circuit **205**. FIGS. **2**G-**2**H illustrate views C and D of FIGS. **2**D-**2**E, respectively, which illustrate details of electrical contacts for electrically coupling transaction card **102** to electric circuit **205**. As illustrated in these figures, card frame **101** may include one or more electrical contacts **211** along the perimeter of the receptacle **103** that contact a corresponding one or more electrical contacts **212** on the perimeter of the transaction card **102** when the transaction card **102** is secured in the receptacle **103**. In FIG. **2**G, contacts **211** and **212** are illustrated on the mating protrusion **208** and groove **207**, respectively, but the contact coupling can be positioned at any location at which the transaction card **102** comes into contact with the card frame **101** so that contacts **211** and **212** touch. Similarly, in FIG. **2**H, contacts **211** and **212** are illustrated on the first rim of card frame **101** and mating second rim of transaction card **102**, respectively, but the contact coupling can be positioned at any location at which the transaction card **102** comes into contact with the card frame **101** so that contacts **211** and **212** touch. For example, if the receptacle **103** is a recess or a slot, the card frame **101** may have contacts on the bottom surface of the recess or inside surface of the slot, which contact corresponding contacts on a surface of the transaction card **102**.

(37) FIGS. **3**A-**3**D illustrate different views of another variation of card assembly **100**, in which card frame **101** communicates with transaction card **102** wirelessly (e.g., without using contacts **211** and **212**). FIG. **3**A illustrates a front view, FIG. **3**B illustrates a back view; and FIGS. **3**C and **3**D illustrate cross-section views C-C of receptacle **300**. As illustrated in these figures, card frame **101** includes an additional antenna **301** proximate to the receptacle **103**. For example, as illustrated in FIGS. **3**A, **3**B, and **3**C, antenna **301** may be embedded in the card frame **101** and may encircle the perimeter of the receptacle **103**, and thus encircle antenna **202** when the transaction card **102** is secured in the receptacle **103**. In other variations, for example when the receptacle **103** comprises a recess or a slot as illustrated in FIG. **3**D, antenna **301** may be embedded in or on the surface of the wall of the slot or on the bottom of a recess. In this assembly, transaction card **102** and card frame **101** exchange data via radio frequency communication between antennas **202** and **301**, which may include implementing a wireless protocol (e.g., NFC, Wi-Fi, Bluetooth®, and/or Bluetooth Low Energy (BLE)). In some variations, antennas **301** and **202** provide inductive power transfer between card frame **101** and transaction card **102**. In some variations the card frame **101** includes both electrical contacts **211** and antenna **301**, which may alternatively be used, or used in combination, depending upon whether the transaction card **102** has corresponding features and capabilities (e.g., contacts **212** and antenna **202**).
(38) Transaction card **102** may be coupled to antenna **201** in the card frame by electric circuit **205** via the electrical contacts **211** and **212** or by the electromagnetically coupled antennas **202** and **301**. In one variation, electric circuit **205** may comprise wire conductors and (optionally) passive components (e.g., capacitors, resistors, inductors) that electrically (e.g., directly or capacitively) connect antenna **201** to contacts **211** and/or antenna **301**.
(39) FIG. **4** illustrates circuit **400**, which is another variation of electric circuit **205**. Circuit **400** may include one or more conductors **405** that are connected between antenna **201** and transceivers and/or amplifiers **401** in card frame **101**. Antenna **201** may receive and radiate radio frequency signals, which correspond to signals carried on the one or more conductors **405** to and from transceivers and/or amplifiers **401**. Similarly, one or more conductors **409** are connected and carry electrical signals between contacts **211** or antenna **301** and transceivers and/or amplifiers **401**. The transceivers and/or amplifiers **401** condition the signals, for example by amplifying and filtering them, and exchange the conditioned signals between conductors **405** and **409** to provide a complete communication path for data carried in the signals between the antenna **201** in the card frame and the transaction card **102** in the receptacle **103**.
(40) For example, the electric circuit **400** may be configured to receive, via contacts **211** and conductors **409**, a first signal comprising transmission data from the transaction card **102** (through contacts **212**), amplify the first signal with transceivers and/or amplifiers **401** to generate an amplified first signal, and transmit wirelessly the amplified first signal including the transmission data via conductors **405** and antenna **201**. Similarly, the electric circuit **400** may be configured to receive wirelessly, via antenna **201** and conductors **405**, a second signal comprising reception data, amplify the second signal to generate an amplified second signal with transceivers and/or amplifiers **401**; and transmit the amplified second signal including the reception data, to the transaction card **102** via conductors **409** and contacts **211**. This relaying of data between the antenna **201** in the card frame **101** and the transaction card **102** in the receptacle **103**, and wirelessly transceiving the data between the antenna **201** and a terminal may be used to perform a contactless data transaction between the transaction card **102** and a terminal. In some variations, transceivers and/or amplifiers **401** may include communication protocol capabilities, such as NFC, Wi-Fi, Bluetooth®, and/or BLE.
(41) Electric circuit **400** may further include a power circuit **403**, which is configured to generate electric power from signals from antenna **201** in the card frame **101**. For example, radio frequency electromagnetic energy (e.g., radio frequency wireless signals) may be received by antenna **201** and conducted along conductors **405** to power circuit **403**. These may be the same or different signals that carry data and are conditioned and amplified by transceivers/amplifiers **401**. Power circuit **403** may include a power converter (for example, comprising a capacitor and a diode) that converts the radio frequency signals to electrical power (e.g., alternating current or direct current power). The generated electrical power may be provided via conductors **407** to energize the electrical circuits within the transceivers and/or amplifiers **401**. The electrical power may additionally or alternatively be provided via conductors **409** to contacts **211** and **212** to the transaction card **102**. In some variations, the electrical power is converted back to radio-frequency signals and transmitted via antenna **301** to the transaction card **102**, which may be configured to receive these signals via antenna **202** and convert them to electrical power internally in the transaction card **102** (e.g., inductive charging).
(42) FIG. **5** illustrates an example method **500** for using the transaction card **102** with and optionally without

card frame **101**. In step **510**, transaction card **102** may be secured in receptacle **103** of card frame **101** as described above (e.g., with an interference fit or magnetic coupling). In step **515**, radio frequency electromagnetic energy (e.g., a wireless radio frequency signal) may be received via antenna **201**.
(43) In step **520**, second antenna **202** in the transaction card **102** is disabled from receiving radio frequency signals from outside of the card frame **101**. In some examples, this prevents the card assembly **100** (**101** and **102** together) from performing or attempting to perform multiple transactions with a card reader (e.g., a point-of-sale terminal), by for example, receiving radio frequency transmission on both antenna **201** and **202**. In some variations (for example, as shown in FIG. **2**F and described above), antenna **202** is disabled by the card frame **101** by providing an electromagnetic shield around the antenna **202**. In other variations, the transaction card **102** detects that it is secured in the receptacle **102**, and based on this detection, disables the antenna **202** internally in the transaction card **102**.
(44) In step **525**, the card frame **101** converts the received radio frequency signals into electrical power (e.g., direct-current or alternating-current power), and in step **530**, the electrical power is provided via electrical contacts **211**/**212** or antennas **301** and **202** (via inductive coupling) to the transaction card **102** in the receptacle **103** as discussed above.
(45) In step **535**, the card frame **101** may relay, via the electrical contacts **211** and **212** or antennas **301** and **202**, electrical signals comprising data between antenna **201** in the card frame **101** and the transaction card **102** in the receptacle **103**. This may be performed by electric circuit **400** as previously discussed, or by computing device **700**, which is further described below with respect to FIG. **7**. In step **540**, card frame **101** may wirelessly transceive this data to and from a card reader **120** (e.g., a terminal). The transceiving may include radiating and/or receiving the data in radio frequency signals from antenna **201**. In step **545**, the card assembly **100** completes a contactless data transaction between the transaction card **102** and the terminal based on the relaying and the transceiving of the data.
(46) In step **550**, the transaction card **102** is removed from the card frame receptacle **103**, as previously described above. Once removed, in step **550**, the transaction card **102** may perform a second contactless data transaction with the card reader (or a different card reader) (e.g., terminal) using its antenna **202**. The steps of process **500** may be performed in other orders and all steps need not be performed.
(47) FIG. **6** illustrates a method **600** for card frame **101**, for example using electric circuit **400**, to relay and transceive data. In step **610**, the card frame **101** may receive (for example via the electrical contacts **211** or antenna **301**) a first signal comprising transmission data from the transaction card **102**. In step **615**, the card frame **101** may amplify (for example using transceivers and/or amplifiers **401**) the first signal to generate an amplified first signal. In step **620**, the card frame **101** may transmit, wirelessly via antenna **201**, the amplified first signal, for example to a card reader **120**. In step **625**, the card frame **101** may receive, via antenna **201**, a second signal comprising reception data, and in step **630**, the card frame **101** may amplify (for example using transceivers and/or amplifiers **401**) the second signal to generate an amplified second signal. In step **635** the card frame **101** may transmit (for example via the electrical contacts **211** or antenna **301**) the amplified second signal to the transaction card **102**. The transmission of the first signal comprising transmission data and the reception of the second signal comprising reception data may be performed in any order and may be related, with one being based on, or in response to, the other, and with both part of a contactless data transaction (e.g., a credit card transaction). Processes **500** and **600** may be performed separately or together.
(48) In some variations of card frame **101**, electric circuit **205** includes a processing circuit for implementing a smart card frame. For example, electric circuit **205** may comprise computing device **700** as illustrated in FIG. **7**. Computing device **700** may include a processor **703** for controlling overall operation of the computing device **700** and its associated components, input/output device **709**, memory **715**, and/or communication interface **723**. A data bus may interconnect processor(s) **703**, memory **715**, I/O device **709**, and/or communication interface **723**.
(49) Input/output (I/O) device **709** may include a port (e.g., contacts, conductors, modem) through which the computing device **700** may receive input, such as for initial programming, receiving authentication keys, etc., prior to being issued to a cardholder.
(50) Software may be stored within memory **715** to provide instructions to processor **703** allowing computing device **700** to perform various actions. For example, memory **715** may store software used by the computing device **700**, such as an operating system **717**, application programs **719**, and/or an associated internal database **721**. The various hardware memory units in memory **715** may include volatile and nonvolatile media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, or other data. Memory **715** may include one or more physical persistent

memory devices and/or one or more non-persistent memory devices. Memory **715** may include, but is not limited to, RAM, ROM, electronically erasable programmable read only memory (EEPROM), flash memory or other memory technology that may store information and that may be accessed by processor **703**.

(51) Communication interface **723** may include one or more transceivers, amplifiers, digital signal processors, and/or additional circuitry and software for communicating via antennas **201** and/or **301** and/or contacts **211**. Communication interface **723** may also include near field communication (NFC) capabilities, short-range wireless communication capabilities (e.g., Bluetooth®), wireless communication capabilities (e.g., Wi-Fi), or any combination thereof. Communication interface **723** may include some or all of the features of electric circuit **400** illustrated in FIG. **4**.

(52) Computing device **700** may further include a power circuit **730**, which may be the same as power circuit **403** described with respect to FIG. **4** for converting radio frequency electromagnetic signals to electrical power for powering computing device **700** and transaction card **102** as previously described.

(53) Processor **703** may include a single central processing unit (CPU), which may be a single-core or multi-core processor, or may include multiple CPUs. Processor(s) **703** and associated components may allow the computing device **700** to execute a series of computer-readable instructions to perform some or all of the processes described herein. Although not illustrated in FIG. **7**, various elements within memory **715** or other components in computing device **700**, may include one or more caches, for example, CPU caches used by the processor **703**, page caches used by the operating system **717**, and/or database caches used to cache content from database **721**. For embodiments including a CPU cache, the CPU cache may be used by one or more processors **703** to reduce memory latency and access time. A processor **703** may retrieve data from or write data to the CPU cache rather than reading/writing to memory **715**, which may improve the speed of these operations.

(54) Although various components of computing device **700** are described separately, functionality of the various components may be combined and/or performed by a single component and/or multiple computing devices in communication. And although various components of computing device **700** are described separately from the various components of electric circuit **400**, these various components and their functionality may be combined and/or performed by a single component and/or multiple computing devices in communication.

(55) The inclusion of a processing circuit, such as computing device **700**, greatly expands the capabilities of card frame **101**, such as: enabling it to work as a transaction card on its own (without transaction card **102** secured in the receptacle), providing security measures limiting the use of the card frame to only certain paired transaction cards **102**, and enabling the card frame to take on multiple different identities, depending upon whether a transaction card is inserted in the receptacle and depending upon which of multiple different transaction cards is inserted into the receptacle.

(56) FIG. **8** illustrates a method **800** for using the card frame **101** that includes a processing circuit, such as computing device **700**. Process **800** begins with step **810**, in which the card frame **101** receives, via antenna **201**, a first communication from a terminal, such as card reader **120**. The first communication may communicate data (in either or both directions) and be the beginning or part of a data transaction (e.g., a contactless transaction, NFC transaction) with the terminal. In step **815** the card frame **101** detects, based on the first communication, whether a transaction card **102** is secured in receptacle **103**. The detection may, for example, be based on a communication between the transaction card **102** and computer device **700**, or may be based on a measurement of an electrical parameter (e.g., detection of a resistance at contacts **211**). In response to detecting that the transaction card **102** is present in the receptacle **103**, steps **820-855** may be performed to complete a contactless data transaction based on the combination of card frame **101** and transaction card **102** together as card assembly **100**. In response to detecting that the transaction card **102** is absent from the receptacle **103**, steps **860-875** may be performed to complete a contactless data transaction based on the card frame **101** alone.

(57) If transaction card **102** is present in the receptacle **103**, card frame **101** in step **820**, performs a second communication with transaction card **102**. The second communication may convey data (in either or both directions) for performing the data transaction with the terminal. The card frame **101** may in step **825** receive in the second communication, a first secure credential from transaction card **102**, which may be unique to transaction card **102**, and may in step **830** authenticate the first secure credential, for example, using a decryption and/or authentication application executed in computing device **700**. In step **835**, the card frame **101** (for example, using computing device **700**) may determine, based on the second communication and/or the authenticated secure credential, an identity of transaction card **102**.

(58) In step **840** the computing device **700** may (optionally) retrieve from a memory (e.g., **715**) in the card frame **101**, a second secure credential uniquely associated with the card frame **101**. Performance of step **840** may be based on confirming that the identity or authenticated secure credential of the transaction card **102** is authorized to be used with the card frame **101**. For example, computing device **700** may have stored in memory, a list of one or more identities of different transaction cards authorized to be used with the card frame **101**. If the transaction card **102** is not authorized to be used with the card frame **101** (e.g., because transaction card **102** is not in the list), the process may end without completing the data transaction.
(59) In step **845** computing device **700** in the card frame **101** may generate a third secure credential based on the first secure credential and (optionally) based on the second secure credential. The third secure credential may be unique to the combination of the card frame **101** and transaction card **102** (for example, by being derived from the first and second secure credentials). Computing device **700** may be configured to generate multiple different third secure credentials based on the second secure credential and, respectively, multiple different first secure credentials of multiple different transaction cards **102**.
(60) In step **850** card frame **101** may perform, via antenna **201** and based on the identity of the transaction card **102** in the receptacle or based the third secure credential, a third communication with the terminal. The performance of steps **845** and/or **850** may be based on or in response to the successful authentication of the first secure credential. The third communication may contain data conveyed in the second communication and additional data (e.g., the third secure credential). In step **855** the card frame **101** may complete a contactless data transaction between the transaction card **102** in the receptacle and the terminal based on data conveyed in the second communication and the third communication. In each of the communications, the data may be conveyed (in either or both directions) and (optionally) encrypted, with computing device **700** performing encryption and decryption of the data.
(61) Returning to step **815**, if the transaction card **102** was determined to be absent from the receptacle **103** integrated in the card frame **101**, step **860** may be performed in which the card frame **101** retrieves from the memory in the card frame **101**, the second secure credential as described above with respect to step **840**. In step **865** card frame **101** may perform, via antenna **201** and based on the second secure credential, a fourth communication with the terminal. In this step, the second secure credential is uniquely associated with just the card frame **101** and is distinct from the secure credentials of the transaction cards **102**. In step **875** card frame **101** completes a data transaction (e.g., contactless data transaction, NFC transaction) between the card frame **101** and the terminal based on data conveyed (in either or both directions) in the fourth communication. In the fourth communication, the data may be encrypted, with computing device **700** performing encryption and decryption of the data.
(62) With the steps of process **800**, the card frame **101** may appear as multiple different transaction cards when performing contactless data transactions, each with a unique identity, that is specific to the card frame **101** alone (with the receptacle **103** empty), or specific to the unique combinations of the card frame **101** and each different transaction card **102** inserted in the receptacle. Moreover, the transaction cards **102** also appear unique with their own respective identities when performing a data transaction without the card frame.
(63) Discussion will now turn to features or characteristics of card assemblies which are paired with a mobile device, such as a smart phone, to provide an additional layer of security when conducting a data transaction with a terminal.
(64) FIG. **9** illustrates a system **90**, which is the same as system **10** illustrated in FIG. **1**, except for the addition of a mobile device **902**, and optionally, the addition of a wireless access point **904** of network **130**. As previously described with respect to system **10**, system **90** may include a card reader **120** (e.g., a point-of-sale terminal), that may exchange data with transaction cards (e.g., **102**) through a plurality of communication techniques. The card reader **120** may be communicatively coupled to a server **140** via network **130**. A network interface may be a wired connection, wireless connection, a short-range wireless connection, a near field communication (NFC) connection, or any combination thereof. The network interface may permit card reader **120** to communicate with server **140**, via network **130**, for example, to authorize a transaction. The network may include wireless access points **904** (one is illustrated for simplicity) enabling devices to wireless connect to the network.
(65) Mobile device **902** may connect to the network via a wireless link **912** to an access point (e.g., **904**) or via another wireless link **914** (e.g., cellular network link). Card reader **120** may also operate as an access point to network **130**.
(66) Server **140** may be a stand-alone server, a corporate server, or a server located in a server farm or cloud-computer environment as previously described to provide cardholders with access to account information and

perform routine banking functions.
(67) Network **130** may be any type of communications and/or computer network. The network **130** may include any type of communication mediums and/or may be based on any type of communication standards or protocols. In this regard, network **130** may include the Internet, a local area network (LAN), a wide area network (WAN), a wireless telecommunications network, and/or any other communication network or a combination thereof.
(68) Devices and systems **120**, **130**, **140**, **902**, and **904** of FIG. **9** may be implemented, in whole or in part, using one or more computing systems, for example, as described below with respect to FIG. **7**.
(69) As previously described, transaction card **102**, card frame **101**, and/or card assembly **100** may include NFC capabilities, short-range wireless communication capabilities (e.g., Bluetooth®), wireless communication capabilities (e.g., Wi-Fi), ultra-wide band (UWB) capabilities, or any combination thereof. The NFC capabilities, short-range wireless communication capabilities, UWB capabilities, and wireless communication capabilities may be referred to collectively as communication capabilities. These communication capabilities may permit transaction card **102**, card frame **101**, and/or card assembly **100** to communicate with card reader **120** via a wireless link **906**, with mobile device **902** via wireless link **908**, or with access point **904** via wireless link **910**.
(70) Using these communication capabilities (and the processing capabilities previously described) transaction card **102**, card frame **101**, and/or card assembly **100** may perform a data transaction according to method **1000** illustrated in FIG. **10**, which may include mobile device **902** being present (or at a known location) to provide an additional level of security. Each of transaction card **102** and card frame **101** includes a computing device (e.g., computing device **700**) to perform the steps of process **1000**. For example, transaction card may include a first memory having a first set of instructions stored therein, and a first processor circuit configured to execute the first set of instructions, that when executed cause the first processor circuit to perform steps of process **1000** as described below. Likewise, card frame **101** may include a second memory having a second set of instructions stored therein, and a second processor circuit configured to execute the second set of instructions, that when executed cause the second processor circuit to perform steps of process **1000** as described below. Certain steps (e.g. steps **1004**A-**1012**A) of process **1000** may be performed by the combination of the first processor and the second processor when transaction card **102** is secured in receptacle **103** of card frame **101**.
(71) Process **1000** begins with step **1002**, in which transaction card **102** or card frame **101** detects whether transaction card **102** is secured in a receptacle **103** of a card frame **101**. For example, the detection may be based on a communication between a computing device (e.g., **700**) in transaction card **102** and a computing device (e.g., **700**) in card frame **101**, or may be based on a measurement of an electrical parameter (e.g., detection of a resistance at contacts **211**) by transaction card **102** or card frame **101** (e.g., as previously described).
(72) If transaction card **102** is detected as being secured in a receptacle **103** (YES), steps **1004**A-**1012**A are performed by card assembly **100** (card frame **101** and transaction card **102**). If transaction card **102** is detected as not being secured in a receptacle **103** (NO), steps **1004**B-**1012**B are performed by transaction card **102** (without card frame **101**).
(73) In step **1004**A (with transaction card **102** secured in the receptacle **103**), or alternatively in step **1004**B (with transaction card not secured in the receptacle **103**), a first credential may be retrieved (e.g., by the first processor) from the first memory in the transaction card **102**. The first credential may be unique to the transaction card **102** and distinguish it from other similar transaction cards compatible with card frame **101**.
(74) In step **1005**A (with transaction card **102** secured in the receptacle **103**), a second credential may be retrieved (e.g., by the second processor) from the second memory in the card frame **101**. The second credential may be unique to the card frame **101** and distinguish it from other similar card frames that may operate with transaction card **102**.
(75) In step **1006**A (with transaction card **102** secured in the receptacle **103**), a third credential may be retrieved, for example, by the first processor from the first memory, or the second processor from the second memory. Alternatively in step **1006**B (with transaction card **102** not secured in the receptacle **103**) the first processor may retrieve the third credential from the first memory. The third credential may be associated with (e.g., uniquely identify) mobile device **902**. When transaction card **102** is secured in receptacle **103**, the first, second, and third credentials may be exchanged between the card frame **101** and the transaction card **102** using the previously described communication methods (e.g., via contacts or antennas) such that either the first processor or the second processor, or both processors may perform steps in process **1000**.

(76) In step **1008**A (with transaction card **102** secured in the receptacle **103**) the card assembly **100** (e.g., using the first processor or the second processor) may determine that mobile device **902** identified by the third credential is within a predetermined distance to the card frame. Alternatively in step **1008**B (with transaction card **102** not secured in the receptacle **103**) transaction card **102** (e.g., the first processer) may determine that mobile device **902** is within a predetermined distance to the transaction card.
(77) There are a number of ways steps **1008**A and **1008**B may be performed. For example, transaction card **102** and/or card frame **101** may include one or more antennas as previously described. Using the one or more antennas, card assembly **100** (e.g. using the first or second processors in step **1008**A), or transaction card **102** (e.g., using the first processor in step **1008**B), may establish a peer-to-peer network (e.g., link **908**) between the mobile device and card frame **101** or transaction card **102**. The peer-to-peer network may utilize a wireless protocol (e.g., ultra-wide band (UWB), NFC, Wi-Fi, Bluetooth®, UWB, and/or Bluetooth Low Energy (BLE)). A maximum distance (e.g., 100 meters) between the peers may be required (depending on the protocol) for the peer-to-peer network to be established. Thus, if the peer-to-peer network is established, a determination may be made that mobile device **902** is at least within a predetermined distance equal to the maximum distance permitted by the communication protocol. The communication protocol being used may also provide an estimated distance (e.g., based the strength of the received signals from the opposite peer), which may be determined or calculated by either peer. From the estimated distance, a determination may be made that mobile device **902** is within a predetermined distance that includes the estimated distance.
(78) Additionally or alternatively, information used in steps **1008**A and **1008**B may be communicated to card frame **101** and/or transaction card **102** via the peer-to-peer network from mobile device **902**, or from network **130** via an access point (e.g., **904**) or card reader (e.g., **120**). For example, mobile device **902** may communicate an estimated distance, coordinates (from a global positioning system) of mobile device **902**, a geographical boundary (e.g., a zip code, city name, distance from a cell phone tower, etc.) that includes mobile device **902**, or an identifier of an access point (e.g., **904**) that mobile device **902** is communicating with (e.g., via link **912**). The card frame **101** and/or transaction card **102** (e.g., using the first processor or the second processer) may have other information about its own location or a geographical boundary it is within, or access point or card reader (e.g., via links **906** and **910**) it is communication with. Using this information, card assembly **100** (e.g. using the first processor or the second processor in step **1008**A), or transaction card (e.g., using the first processor in step **1008**B), may determine that the mobile device **902** is within the predetermined distance. For example, the distance between card assembly **100**/transaction card **102** and mobile device **902** may be determined by comparing geographical boundaries, calculating distance between respective locations (e.g., GPS coordinates), determining whether the card assembly **100**/transaction card **102** and mobile device **902** are communicating with the same access point or communicating with different access points that within a known distance or geographical boundary, etc.
(79) In other examples, a separate device or system, such as mobile device **902** or server **140** may determine that card assembly **100**/transaction card **102** and mobile device **902** are at a particular distance or within the predetermined distance using the same information. The particular distance (or the determination that the devices are within the predetermined distance), may then be communicated to card assembly **100** or transaction card **102** via network **130** and the card reader **120** (via link **906**), mobile device **902** (via link **908**), or access point **904** (via link **910**).
(80) The information communicated to card assembly **100** or transaction card **102** may include the third credential to indicate that the associated mobile device **902** is within the predetermined distance. Different mobile devices may be used in process **1000**, each with a different third credential communicated to card assembly **100** or transaction card **102**. The third credential of each of the different mobile devices may be stored in the first memory or the second memory (e.g., after it is communicated to the card assembly **100** or transaction cared **102**), and recalled in steps **1006**A or **1006**B (e.g., in a subsequent performance of process **1000**).
(81) In step **1010**A (with transaction card **102** secured in the receptacle **103**) and step **1010**B (with transaction card **102** not secured in the receptacle **103**), a new credential (e.g., an authentication credential) is generated by card assembly **100** or transaction card **102**, respectively. The generation of the new credential may be based on the previous determination in steps **1008**A and **1008**B that mobile device **902** is within the predetermined distance to the card assembly **100** or transaction card **102**.
(82) In step **1010**A, the first processor in transaction card **102** or the second processor in card frame **101** (or both processors) may generate the new credential from the first credential associated with the transaction card **102**, the second credential associated the card frame **101**, and the third credential associated with the mobile

device **902**. In step **1010**B, the first processor in transaction card **102** may generate the new credential from the first credential and the third credential. The first processor in transaction card **102** and the second processor in card frame **101** may each (or alternatively) include an authentication code generator in which the first credential, second credential (in step **1010**A), and third credential may be used as seed inputs for generating an authentication credential.
(83) In some variations, based on the same first, second (in step **1010**A), and third credentials, the code generator may generate a new (e.g., authentication) credential that is different than the previously generated credential each time steps **1010**A or **1010**B is performed. For example, the generated credential may be based on the first, second (in step **1010**A), and third credentials, and a moving factor as seed inputs. The moving factor may be hash based. For example, generation of the hash based moving factor may utilize the output of an incremental counter that advances each time the generator produces a new credential. Alternatively, or additionally, the moving factor may be based on time. For example a time stamp of the current time, an elapsed time since the generation of the last credential, or an incremental count that advances at predetermined intervals (e.g., 18 seconds) may be used as a seed to the code generator. The moving factor may be calculated or determined (e.g., by incrementing a counter or maintaining a clock) by the first processor in transaction card **102** or by the second processor in card frame **101**. Alternatively, the moving factor (e.g., a count or a time stamp) may be generated externally (e.g., by server **140**, mobile device **902**, access point **904**, or card reader **120**) and communicated to card assembly **100** or transaction card **102** via one of the communication links (e.g., **906**, **908**, **910**). Using the moving factor, a one-time authentication credential may be generated each time steps **1010**A or **1010**B is performed.
(84) In step **1012**A (with transaction card **102** secured in the receptacle **103**) and in step **1012**B (with transaction card **102** not secured in the receptacle **103**), the transaction card performs a data transaction between the transaction card and terminal **120** (or one of a plurality of terminals) using the credential generated in steps **1010**A and **1010**B, respectively. For example, a data transaction may be performed as previously describe with respect process **500** (FIG. **5**) or process **800** (FIG. **8**), with the generated credential being used to authenticate the transaction card **102** (e.g., indicating the card is the originally issued credit card and in possession of an authorized user).
(85) Process **1000** may be performed multiple times with a plurality of card frames securing the same transaction card, or a plurality of transaction cards being secured in the same card frame. Moreover, process **1000** may be performed with each transaction card, or different combination of card frame and transaction card in conjunction with a plurality of mobile devices. In each performance, each of the plurality of card frames, each of the plurality of transaction cards, and each of the plurality of mobile devices respectively have one of a plurality of credentials stored therein. Each of the plurality of credentials may be unique, such that each combination of a first credential (from one of the plurality of transaction cards), a third credential (from one of the plurality of mobile devices), and (optionally) a second credential (from one of the plurality of card frames) generates a unique credential (e.g., authentication credential) for performing respective data transactions between the transaction cards and terminal **120** (or one of a plurality of terminals).
(86) One or more aspects discussed herein (e.g., processes and functions being performed by a processor or computer chip as described above) may be embodied in computer-usable or readable data and/or computer-executable instructions, such as in one or more program modules, executed by one or more computers or other devices as described herein. Generally, program modules include routines, programs, objects, components, and data structures that perform particular tasks or implement particular abstract data types when executed by a processor in a computer or other device. The modules may be written in a source code programming language that is subsequently compiled for execution, or may be written in a scripting language such as (but not limited to) HTML or XML. The computer executable instructions may be stored on a computer readable medium such as solid-state memory, RAM, and the like. As will be appreciated by one of skill in the art, the functionality of the program modules may be combined or distributed as desired in various embodiments. In addition, the functionality may be embodied in whole or in part in firmware or hardware equivalents such as integrated circuits, field programmable gate arrays (FPGA), and the like. Particular data structures may be used to more effectively implement one or more aspects discussed herein, and such data structures are contemplated within the scope of computer executable instructions and computer-usable data described herein. Various aspects discussed herein may be embodied as a method, a computing device, a system, and/or a computer program product.
(87) Although the present invention has been described in certain specific aspects, many additional modifications and variations would be apparent to those skilled in the art. In particular, any of the various

processes described above may be performed in alternative sequences and/or in parallel (on different computing devices) in order to achieve similar results in a manner that is more appropriate to the requirements of a specific application. It is therefore to be understood that the present invention may be practiced otherwise than specifically described. Thus, embodiments of the present disclosure should be considered in all respects as illustrative and not restrictive.

Claims

1. A transaction card comprising: a memory having instructions stored therein; and a processor circuit configured to execute the instructions, and based on execution of the instructions, to: determine that a mobile device is within a predetermined distance to the transaction card; perform, based on the determination that the mobile device is within the predetermined distance to the transaction card, a secure data transaction between the transaction card and a terminal; receive, from the terminal or the mobile device, a first credential associated with the mobile device; retrieve, from the memory, a second credential; generate a third credential based on the first credential and the second credential; and perform the secure data transaction further based on the first credential, the second credential, and the third credential.
2. The transaction card of claim 1, wherein the processor circuit is configured, based on the execution of the instructions, to: receive, from the terminal, information about the mobile device, wherein the determination that the mobile device is within the predetermined distance to the transaction card is based on the information.
3. The transaction card of claim 1, wherein the processor circuit is configured, based on the execution of the instructions, to: determine a second occurrence of the mobile device being within the predetermined distance to the transaction card; generate, based the determination of the second occurrence of the mobile device being within the predetermined distance to the transaction card, a fourth credential based on the first credential and the second credential, wherein the fourth credential differs from the third credential; and perform, using the fourth credential, a second secure data transaction between the transaction card and: the terminal or a second terminal.
4. The transaction card of claim 1, wherein the processor circuit is configured, based on the execution of the instructions, to: perform the secure data transaction further based on a moving factor.
5. The transaction card of claim 1, further comprising an antenna, wherein the processor circuit is configured, based on the execution of the instructions, to: receive, via the antenna, information from the mobile device, wherein the determination that the mobile device is within the predetermined distance is based on the information.
6. The transaction card of claim 5, wherein the processor circuit is configured, based on the execution of the instructions, to: retrieve, from the memory, a credential associated with the transaction card; extract a moving factor from the information; and perform the secure data transaction based on the credential and the moving factor.
7. The transaction card of claim 1, wherein the processor circuit is configured, based on the execution of the instructions, to: detect that the transaction card is secured in a receptacle of a card frame; and perform the secure data transaction further based on the detection that the transaction card is secured in the receptacle.
8. A transaction card comprising: a memory having instructions stored therein; and a processor circuit configured to execute the instructions, and based on execution of the instructions, to: receive, from a terminal or a mobile device, a first credential associated with the mobile device; retrieve, from the memory, a second credential associated with the transaction card; perform, based on the first credential and the second credential, a secure data transaction between the transaction card and the terminal; and generate a third credential based on the first credential and the second credential, wherein the secure data transaction includes a transmission of the third credential from the transaction card to the terminal.
9. The transaction card of claim 8, wherein the processor circuit is configured, based on the execution of the instructions, to: generate a fourth credential based on the first credential and the second credential; and perform a second secure data transaction between the transaction card and the terminal or a second terminal, wherein the second secure data transaction includes a second transmission of the fourth credential from the transaction card to the terminal or the second terminal, and wherein the third credential and the fourth credential are different.
10. The transaction card of claim 8, wherein the processor circuit is configured, based on the execution of the instructions, to: generate a moving factor based on time or a count; and perform the secure data transaction further based on the moving factor.

11. The transaction card of claim 8, further comprising an antenna, wherein the processor circuit is configured, based on the execution of the instructions, to: receive the first credential via the antenna.
12. The transaction card of claim 11, wherein the processor circuit is configured, based on the execution of the instructions, to: receive the first credential directly from the mobile device.
13. The transaction card of claim 8, wherein the processor circuit is configured, based on the execution of the instructions, to: receive, from the terminal, an indication that the mobile device is within a predetermined distance to the transaction card; and perform the secure data transaction based on the indication.
14. The transaction card of claim 8, wherein the processor circuit is configured, based on the execution of the instructions, to: detect that the transaction card is secured in a receptacle of a card frame; and perform the secure data transaction further based on the detection that the transaction card is secured in the receptacle.
15. A method comprising: determining that a mobile device is within a predetermined distance to a transaction card; performing, based on the determining that the mobile device is within the predetermined distance, a secure data transaction between the transaction card and a terminal; receiving, by the transaction card from the terminal or the mobile device, a first credential associated with the mobile device; retrieving, from a memory in the transaction card, a second credential; generating a third credential based on the first credential and the second credential; and perform the secure data transaction further based on the first credential, the second credential, and the third credential.
16. The method of claim 15, further comprising: receiving, by the transaction card from the terminal, information about the mobile device, wherein the determining that the mobile device is within the predetermined distance is based on the information.
17. The method of claim 15, further comprising: performing, via an antenna in the transaction card, a peer-to-peer network communication with the mobile device, wherein the determining that the mobile device is within the predetermined distance is based on the peer-to-peer network communication.
18. The method of claim 15, further comprising: detecting that the transaction card is secured in a receptacle of a card frame; and performing the secure data transaction further based on the detection that the transaction card is secured in the receptacle.
19. The method of claim 15, further comprising: generate a fourth credential based on the first credential and the second credential; and perform a second secure data transaction between the transaction card and the terminal or a second terminal, wherein the second secure data transaction includes a second transmission of the fourth credential from the transaction card to the terminal or the second terminal, and wherein the third credential and the fourth credential are different.
20. The method of claim 15, further comprising: generating a moving factor based on time or a count; and performing the secure data transaction further based on the moving factor.